MATHEMATICS SECTION 1 (Maximum Marks : 12)

- This section contains **FOUR (04)** questions.
- Each question has **FOUR options** (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated **according to the following marking scheme:**

  Full Marks : +3 If **ONLY** the correct option is chosen
  Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
  Negative Marks : -1 In all other cases.

Q.1 Let $H(x)$ be a function defined on the interval $(0, 2]$ as:

$$H(x) = (x^2 - 2)(3x - 4)\left(\left[\frac{x^2}{2}\right] + \left[\frac{3x}{2}\right] + [2x]\right)$$

(where $[\cdot]$ denotes the greatest integer function).

Let Set A contain all values of x in $(0, 2]$ where $H(x)$ is **continuous but non-differentiable**.
Let Set B contain all values of x in $(0, 2]$ where $H(x)$ is **discontinuous**.

Which of the following correctly defines Set A and Set B?

(A) $A = \left\{\frac{4}{3}, \sqrt{2}\right\}, \quad B = \left\{\frac{1}{2}, \frac{2}{3}, 1, \frac{3}{2}, 2\right\}$	**(B)** $A = \emptyset, \quad B = \left\{\frac{1}{2}, \frac{2}{3}, 1, \frac{4}{3}, \sqrt{2}, \frac{3}{2}, 2\right\}$
(C) $A = \left\{\frac{4}{3}, \sqrt{2}, 2\right\}, \quad B = \left\{\frac{1}{2}, \frac{2}{3}, 1, \frac{3}{2}\right\}$	**(D)** $A = \left\{\frac{1}{2}, 1, \frac{3}{2}\right\}, \quad B = \left\{\frac{2}{3}, \frac{4}{3}, \sqrt{2}, 2\right\}$

Answer: A

Q.2 Let $A(x)$ and $B(x)$ be 3×3 matrices defined as:

$$A(x) = \begin{pmatrix} x & 2 & a_1 \\ 0 & x & 3 \\ a_2 & 0 & x \end{pmatrix} \quad \text{and} \quad B(x) = \begin{pmatrix} x & 5 & b_1 \\ 0 & x & 1 \\ b_2 & 0 & x \end{pmatrix}$$

where $a_1, a_2, b_1, b_2 \in \mathbb{R}$ and $x \in \mathbb{R}$.

Define $C(x) = \det(A(x+1)) - \det(B(x-1))$.

If $\det(A(x)) \neq \det(B(x))$ for every real number x, what is the coefficient of x^2 in $C(x)$?

(A) 0	**(B)** 6
(C) 9	**(D)** -3

Answer: B

Q.3 Let S be a set containing the first 8 natural numbers, i.e., $S = \{1, 2, 3, 4, 5, 6, 7, 8\}$. Two subsets A and B of S are chosen independently and at random. It is assumed that all possible subsets of S are equally likely to be selected for both A and B. Given that the union of the selected subsets A and B contains exactly 5 elements, what is the probability that subset A is completely contained within subset B (i.e., $A \subseteq B$)?

(A) $\frac{32}{243}$	**(B)** $\frac{1}{8}$
(C) $\frac{16}{243}$	**(D)** $\frac{1}{243}$

Answer: A

Q.4 Let α and β be two acute angles such that $0 < \alpha, \beta < \frac{\pi}{2}$.

Consider the following trigonometric equation:
$\log_{\cos(\alpha)}(\sin(\alpha)) + \log_{\sin(\alpha)}(\cos(\alpha)) = 2$

Let θ be the largest angle of a triangle whose side lengths are $3, 5$, and 7.

If β satisfies the equation $\sin(\beta) = \sin(\alpha)\cos\left(\frac{\theta}{4}\right) + \cos(\alpha)\sin\left(\frac{\theta}{4}\right)$, then which of the following is the value of the expression $\frac{12\beta}{\alpha}$?

(A) 15	**(B)** 18
(C) 20	**(D)** 24

Answer: C

MATHEMATICS SECTION 2 (Maximum Marks : 12)

- This section contains **THREE (03)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated **according to the following marking scheme:**

  Full Marks : +4 **ONLY** if (all) the correct option(s) is(are) chosen;
  Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen;
  Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct;
  Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option;
  Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
  Negative Marks : -2 In all other cases.

Q.5 Let C_1 be the locus of z satisfying $|z-3-4i|=5$ and C_2 be the locus of z satisfying $|z-3+4i|=5$. Let Z_0 be the non-zero point of intersection of C_1 and C_2.

Let L be the straight line passing through Z_0 and the origin $O(0,0)$. Let a new circle C_3 have its center at $P(3+4i)$ and be tangent to the line L at point Q.

Let R be the point on C_3 that is at a maximum distance from the origin.

Then which of the following statements is (are) TRUE?

(A) The principal argument of the complex number represented by R is $\tan^{-1}\left(\frac{3}{4}\right)$	**(B)** The length of the line segment PQ is 4
(C) The area of $\triangle OPQ$ is 6	**(D)** The square of the length of the line segment QR is $\frac{288}{5}$

Answer: B, C, D

Q.6

Let $f(r)$ be the number of distinct common tangents between the parabola $y^2 = 4x$ and the circle $(x-3)^2 + y^2 = r^2$, defined for all real $r > 0$.

Let $g(k)$ be the number of distinct real normals that can be drawn from the point $(k, 0)$ to the parabola $y^2 = 4x$, defined for all real k.

Define the composite functions $(g \circ f)(r) = g(f(r))$ and $(f \circ g)(k) = f(g(k))$.

Which of the following statements is (are) TRUE?

(A) The range of the function $f \circ g$ is exactly $\{0, 3\}$.	**(B)** $(g \circ f)(r) = 3$ for all $r \in (2\sqrt{2}, 3]$.
(C) $f(r)$ is continuous for all $r > 0$.	**(D)** The set of values of r satisfying $(g \circ f)(r) = 1$ consists of a finite number of discrete points.

Answer: A, B

Q.7

Let $\mathbb{R}^3$ denote the three-dimensional space. Let $a = 2\hat{i} - \hat{j} + 2\hat{k}$ and $b = \hat{i} + 3\hat{j} - \hat{k}$ be two constant position vectors.

Let $S = \left\{ r = x\hat{i} + y\hat{j} + z\hat{k} \in \mathbb{R}^3 : (r - a) \cdot (r - b) = |r|^2 - 24 \right\}$

Then which of the following statements is (are) TRUE?

(A) The perpendicular distance from the origin to S is $\dfrac{3\sqrt{14}}{2}$	**(B)** The intercepts made by S on the $x, y,$ and z axes are $7, 21,$ and $\dfrac{21}{2}$ respectively
(C) S contains the point $(2, 5, 4)$	**(D)** The normal vector to S is parallel to the vector $6\hat{i} + 4\hat{j} + 2\hat{k}$

Answer: A, D

MATHEMATICS SECTION 3 (Maximum Marks : 24)

- This section contains **SIX (06)** questions.
- The answer to each question is a **NUMERICAL VALUE.**
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated **according to the following marking scheme:**

  Full Marks : +4 If **ONLY** the numerical answer is in acceptable range
  Zero Marks : 0 In all other cases

Q.8 Let $\mathcal{S}$ be the set of all equivalence relations R on the set $A = \{1, 2, 3, 4, 5\}$ such that R contains exactly 11 elements.

Then the number of elements in $\mathcal{S}$ is ____.

Answer: 10

Q.9 Let $f(x)$ be a continuous, strictly positive function on $\mathbb{R}$. For any two real numbers a and b, let I_a^b denote the definite integral $\int_a^b f(x)\, dx$.

Let p, q, and r be three distinct real numbers. Let s be a real number such that:

$I_s^p + 5I_s^q + 6I_s^r = 0$

Let u and v be real numbers such that $I_p^u = I_u^r$ and $I_q^v = I_v^r$.

Then the value of $\left|\frac{I_u^v}{I_u^s}\right|$ is ____.

Answer: 1.2

Q.10 Let S be the set of all vectors $v = x\hat{i} + y\hat{j} + z\hat{k}$, where the components x, y, and z are integers such that $-10 \le x, y, z \le 10$.

Let $n = \hat{i} + \hat{j} + \hat{k}$. A vector v in S is considered "valid" if it is **NOT** orthogonal to n (for example, $v = 5\hat{i} - 2\hat{j} - 3\hat{k}$ is NOT valid).

Find the number of valid vectors in S such that the vector's x and y components are equal, **OR** its y and z components are equal.

Answer: 840

Q.11 Let α and β be real numbers. Let $y = y(x)$ be the solution to the differential equation

$$(1+x^2)\frac{dy}{dx} + 2xy = \alpha x e^x + \beta \sin x$$

subject to the initial condition $y(0) = 1$.

If $\lim_{x\to 0} \dfrac{y(x)-1}{x^3} = 1$, then find the value of $\alpha - \beta$.

Answer: 4

Q.12 Let $\mathbb{C}$ denote the set of all complex numbers. Let $f : \mathbb{R} \to \mathbb{C}$ be a function such that $f(x) \neq 0$ for all $x \in \mathbb{R}$, and $f(x+y) = f(x)f(y)$ for all $x, y \in \mathbb{R}$.

Let the real numbers $a_1, a_2, \ldots, a_{50}$ be in an arithmetic progression with common difference d. Suppose $f(a_{31}) = -64 f(a_{25})$, $\operatorname{Re}(f(d)) = 0$, and $\operatorname{Im}(f(d)) > 0$. If

$\sum_{k=1}^{50} f(a_k) = 3(2^{25} - i)$

then the value of

$\sum_{k=6}^{30} f(a_k)$

is _______.

Answer: 96

Q.13 Let α and β be non-zero real numbers. For a real number $t \neq 0$, let $P(t, \alpha t^2 + \beta t^4)$ and $Q(-t, \alpha t^2 + \beta t^4)$ be two points on a curve. Let C_t be the unique circle passing through P, Q, and the origin $O(0,0)$. Let the y -coordinate of the center of C_t be denoted by $y_c(t)$.

If

$\lim_{t\to 0} \dfrac{y_c(t) - 1}{t^2} = 2$

Then the value of $4\alpha - 16\beta$ is ____.

Answer: 16

MATHEMATICS SECTION 4 (Maximum Marks : 12)

- This section contains **FOUR (04)** Matching List Sets.
- Each set has **ONE** Multiple Choice Question.
- Each set has **TWO** lists: **List-I** and **List-II**.
- **List-I** has **Four** entries (P), (Q), (R) and (S) and **List-II** has **Five** entries (1), (2), (3), (4) and (5).
- **FOUR** options are given in each Multiple Choice Question based on **List-I** and **List-II** and **ONLY ONE** of these four options satisfies the condition asked in the Multiple Choice Question.
- Answer to each question will be evaluated **according to the following marking scheme:**

  Full Marks : +3 **ONLY** if the option corresponding to the correct combination is chosen;
  Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
  Negative Marks : -1 In all other cases

Q.14 Let a, b, and c be the three distinct real roots of the cubic equation $x^3 - 3x + 1 = 0$.

Let $S_n = a^n + b^n + c^n$ for any integer $n \geq 0$.

Match each entry in List-I to the correct entry in List-II.

List-I	List-II
(P) The value of S_4 is	(1) 1
(Q) The value of $-(a^5 + b^5 + c^5)$ is	(2) 15
(R) Let $Q(x) = (x + a + b)(x + b + c)(x + c + a)$. The absolute value of $Q(1)$ is	(3) 17
(S) The value of $(a^2 + 1)(b^2 + 1)(c^2 + 1)$ is	(4) 18
	(5) 24

(A) $(P) \to (4)$ $(Q) \to (3)$ $(R) \to (2)$ $(S) \to (1)$	**(B)** $(P) \to (4)$ $(Q) \to (2)$ $(R) \to (1)$ $(S) \to (5)$
(C) $(P) \to (2)$ $(Q) \to (3)$ $(R) \to (5)$ $(S) \to (1)$	**(D)** $(P) \to (4)$ $(Q) \to (2)$ $(R) \to (1)$ $(S) \to (3)$

Answer: D

Q.15

Class Interval	$1-3$	$4-6$	$7-9$	$10-12$	$13-15$
Frequency	3	f_1	8	f_2	2

Suppose that the sum of the frequencies is 25 and the median of this frequency distribution is $\frac{131}{16}$ (or 8.1875).

For the given frequency distribution, let α denote the mean deviation about the mean, σ^2 denote the variance, and M_o denote the mode.

Match each entry in List-I to the correct entry in List-II and choose the correct option.

List-I	**List-II**
(P) $f_1^2+f_2^2$ is equal to	(1) 66
(Q) 25α is equal to	(2) 288
(R) $25\sigma^2$ is equal to	(3) 74
(S) $4M_o$ is equal to	(4) 35
	(5) 48

(A) $(P)\to(3),(Q)\to(4),(R)\to(5),(S)\to(1)$	**(B)** $(P)\to(1),(Q)\to(3),(R)\to(2),(S)\to(5)$
(C) $(P)\to(3),(Q)\to(1),(R)\to(2),(S)\to(4)$	**(D)** $(P)\to(3),(Q)\to(1),(R)\to(5),(S)\to(2)$

Answer: C

Q.16

Let an ellipse be given by $E : \frac{x^2}{16} + \frac{y^2}{9} = 1$. A circle S with center $(0, \alpha)$, where $\alpha > 0$, and radius r touches the ellipse E at two distinct points P and Q. It is given that the length of the line segment PQ is $2\sqrt{7}$.

Match each entry in Column I to the correct entry in Column II.

Column I	Column II
(P) The value of 4α equals	(1) 4
(Q) The value of r^2 equals	(2) 7
(R) If the tangents drawn to E at P and Q intersect at a point $T(0, y_0)$, then $\lvert y_0 \rvert$ equals	(3) 16
(S) The value of $16e^2$, where e is the eccentricity of E, equals	(4) 23
	(5) 25

(A) $(P) \rightarrow (2), (Q) \rightarrow (4), (R) \rightarrow (1), (S) \rightarrow (2)$	**(B)** $(P) \rightarrow (1), (Q) \rightarrow (5), (R) \rightarrow (3), (S) \rightarrow (4)$
(C) $(P) \rightarrow (1), (Q) \rightarrow (5), (R) \rightarrow (2), (S) \rightarrow (3)$	**(D)** $(P) \rightarrow (2), (Q) \rightarrow (4), (R) \rightarrow (3), (S) \rightarrow (1)$

Answer: A

Q.17 Let $\mathbb{R}$ denote the set of all real numbers. Match each entry in List-I to the correct entry in List-II and choose the correct option.

List-I	List-II
(P) Let $f(x)$ be a polynomial of degree 4 with local extrema at $x = 1$ and $x = 2$. If $\lim_{x \to 0} \left(1 + \frac{f(x)}{x^2}\right) = 5$, then the number of distinct real roots of the equation $f(x) = \frac{1}{2}$ is	(1) 2
(Q) The number of points in the interval $(0, 2\pi)$ where the function $g(x) = \|\sin 2x\| + \|\cos x\|$ is NOT differentiable, is	(2) 3
(R) Let d be the shortest distance between the parabola $y = x^2$ and the line $y = x - 2$. The value of $4\sqrt{2}d$ is	(3) 4
(S) Let $h(x) = \frac{K}{3}x^3 - Kx^2 + 4x + 2$. The number of integers K for which $h(x)$ is strictly increasing on $\mathbb{R}$ is	(4) 5
	(5) 7

(A) $(P) \to (4), (Q) \to (5), (R) \to (1), (S) \to (2)$	**(B)** $(P) \to (3), (Q) \to (2), (R) \to (5), (S) \to (4)$
(C) $(P) \to (4), (Q) \to (2), (R) \to (1), (S) \to (3)$	**(D)** $(P) \to (3), (Q) \to (5), (R) \to (4), (S) \to (1)$

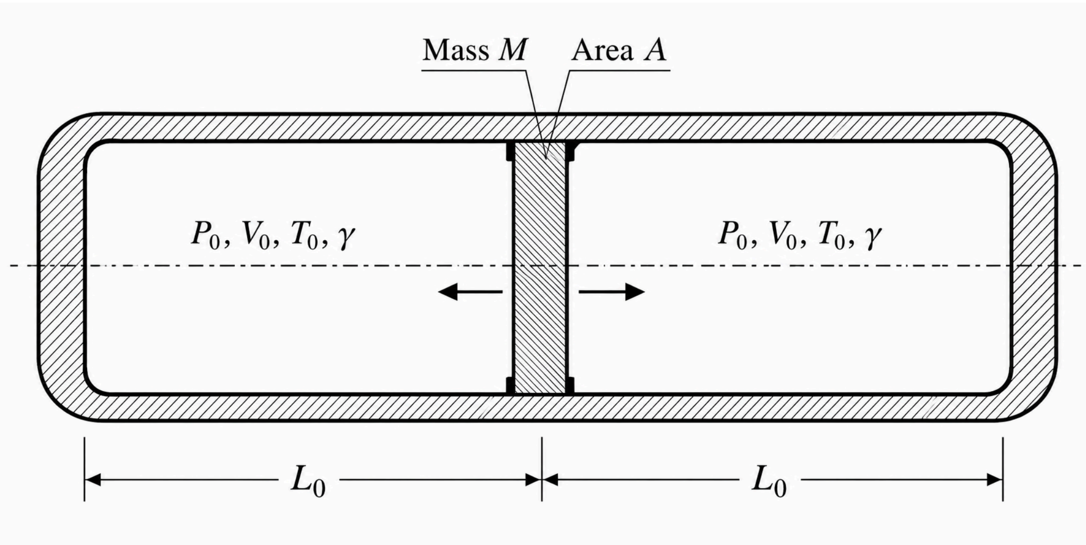
Answer: B

PHYSICS SECTION 1 (Maximum Marks : 12)

- This section contains **FOUR (04)** questions.
- Each question has **FOUR options** (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated **according to the following marking scheme:**

  Full Marks : +3 If **ONLY** the correct option is chosen
  Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
  Negative Marks : -1 In all other cases.

Q.18



A long, rigid, insulated horizontal cylinder is divided into two identical compartments, each of volume V_0 and pressure P_0, by a heavy, insulating piston of mass M and cross-sectional area A. The initial temperature of the ideal gas (with known adiabatic index γ) in each compartment is T_0. The entire cylinder is stationary.

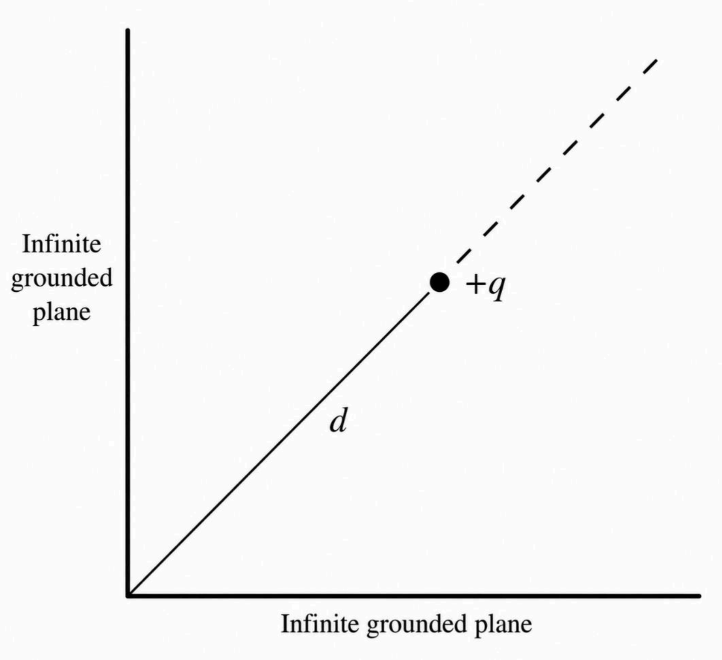
The piston is then given a small horizontal displacement $x \ll V_0/A$ and released, with both compartments undergoing adiabatic processes.

The correct expression for the angular frequency ω of these small oscillations of the piston is given by:

(A) $\sqrt{\dfrac{\gamma P_0 A^2}{M V_0}}$	**(B)** $\sqrt{\dfrac{2 P_0 A^2}{M V_0}}$
(C) $\sqrt{\dfrac{2\gamma P_0 A^2}{M V_0}}$	**(D)** $\sqrt{\dfrac{P_0 A^2}{M V_0}}$

Answer: C

Q.19



Two infinite, grounded, conducting planes intersect at an angle of 90°. A point charge $+q$ is placed on the angle bisector at a distance d from the intersection line.

What is the total electrostatic potential energy of this system?

(A) $-\frac{q^2}{4\pi\varepsilon_0 d}\left(\sqrt{2}-\frac{1}{2}\right)$	**(B)** $-\frac{q^2}{8\pi\varepsilon_0 d}\left(\sqrt{2}-\frac{1}{2}\right)$
(C) $-\frac{q^2}{4\pi\varepsilon_0 d}\left(1-\frac{1}{2\sqrt{2}}\right)$	**(D)** $-\frac{q^2}{8\pi\varepsilon_0 d\sqrt{2}}$

Answer: B

Q.20 The velocity v of surface waves on a liquid is given by the empirical formula $v = k\lambda^a \rho^b S^c$, where λ is wavelength, ρ is density, S is surface tension, and k is a dimensionless constant.

A student determines the surface tension S using the capillary rise method formula $S = \dfrac{rh\rho g}{2}$. The height h of the liquid column is found by taking the difference between two independent readings on a vertical scale: an upper meniscus reading of $6.0\ \mathrm{cm}$ and a lower reservoir reading of $2.0\ \mathrm{cm}$. The scale has a least count of $0.1\ \mathrm{cm}$, and the absolute error in each individual scale reading is equal to this least count.

The percentage errors in the other measured quantities are:
Radius of capillary, $r = 2\%$
Wavelength, $\lambda = 4\%$
Density, $\rho = 3\%$

Assume g and k are exact constants. What is the true maximum percentage error in the calculated velocity v ?

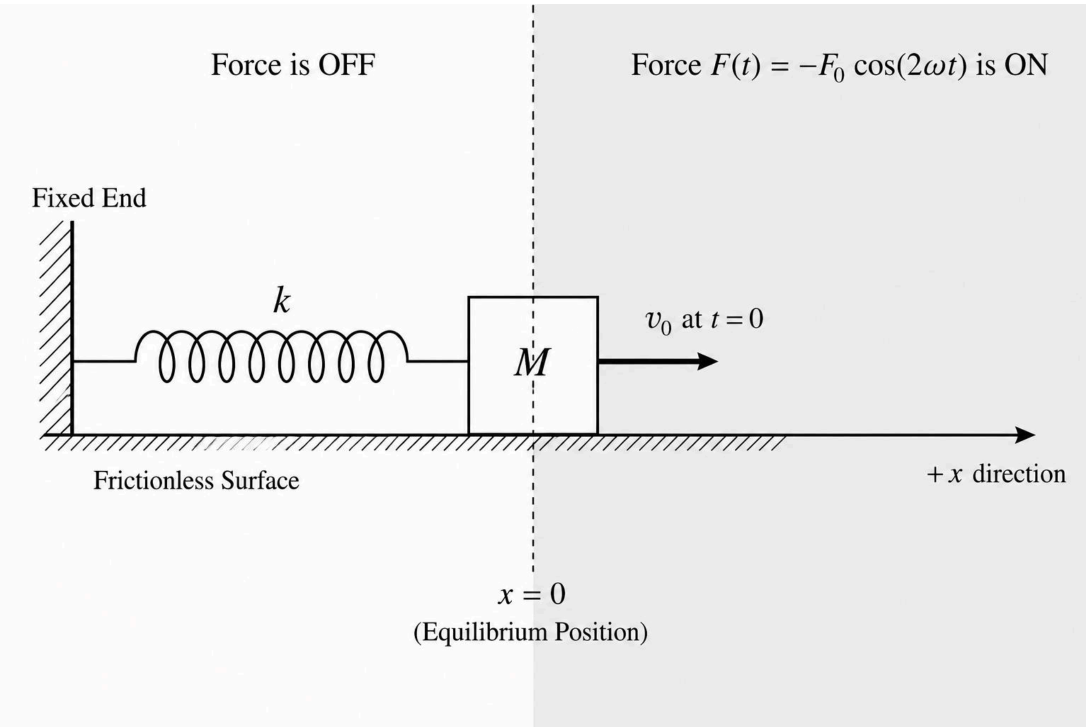
(A) 4.25%	**(B)** 5.50%
(C) 7.25%	**(D)** 8.50%

Answer: B

Q.21 A block of mass M is attached to a horizontal ideal spring of constant k (where natural frequency $\omega = \sqrt{k/M}$). The block rests on a frictionless surface with equilibrium at $x = 0$. At $t = 0$, the block is projected from $x = 0$ with an initial velocity $+v_0\hat{i}$. An external time-dependent force $F(t) = -F_0\cos(2\omega t)$ is applied to the block, but it only acts in the region $x \geq 0$. The external force is zero everywhere else.

Which of the following expressions correctly represents the external force $F(t)$ experienced by the block during its first complete cycle of oscillation?

Assume the magnitude of the external force F_0 is extremely small compared to the spring force, such that the period of oscillation is approximately unchanged.



(A) $$F(t) = \begin{cases} -F_0\cos(2\omega t) & \text{for } t \in [0, \pi/\omega] \\ 0 & \text{for } t \in (\pi/\omega, 2\pi/\omega] \end{cases}$$

(B) $$F(t) = \begin{cases} -F_0\cos(\omega t) & \text{for } t \in [0, \pi/\omega] \\ 0 & \text{for } t \in (\pi/\omega, 2\pi/\omega] \end{cases}$$

(C) $F(t) = -F_0\cos(2\omega t)$ for all $t > 0$

(D) $$F(t) = \begin{cases} 0 & \text{for } t \in [0, \pi/2\omega) \\ -F_0\cos(2\omega t) & \text{for } t \in [\pi/2\omega, 3\pi/2\omega] \\ 0 & \text{for } t \in (3\pi/2\omega, 2\pi/\omega] \end{cases}$$

Answer: A

PHYSICS SECTION 2 (Maximum Marks : 12)

- This section contains **THREE (03)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated **according to the following marking scheme:**

  Full Marks : +4 **ONLY** if (all) the correct option(s) is(are) chosen;
  Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen;
  Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct;
  Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option;
  Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
  Negative Marks : -2 In all other cases.

Q.22 The first excitation potential of a single-electron ion is given as 4.08×10^4 mV. A macroscopic sample of these ions contains electrons in an excited state where their orbital kinetic energy is 5.44×10^{-19} J. As these ions completely de-excite to the ground state, which of the following option(s) is/are correct?

(Take $hc = 1240 \text{ eV} \cdot \text{nm}$, $1 \text{ eV} = 1.6 \times 10^{-19}$ J)

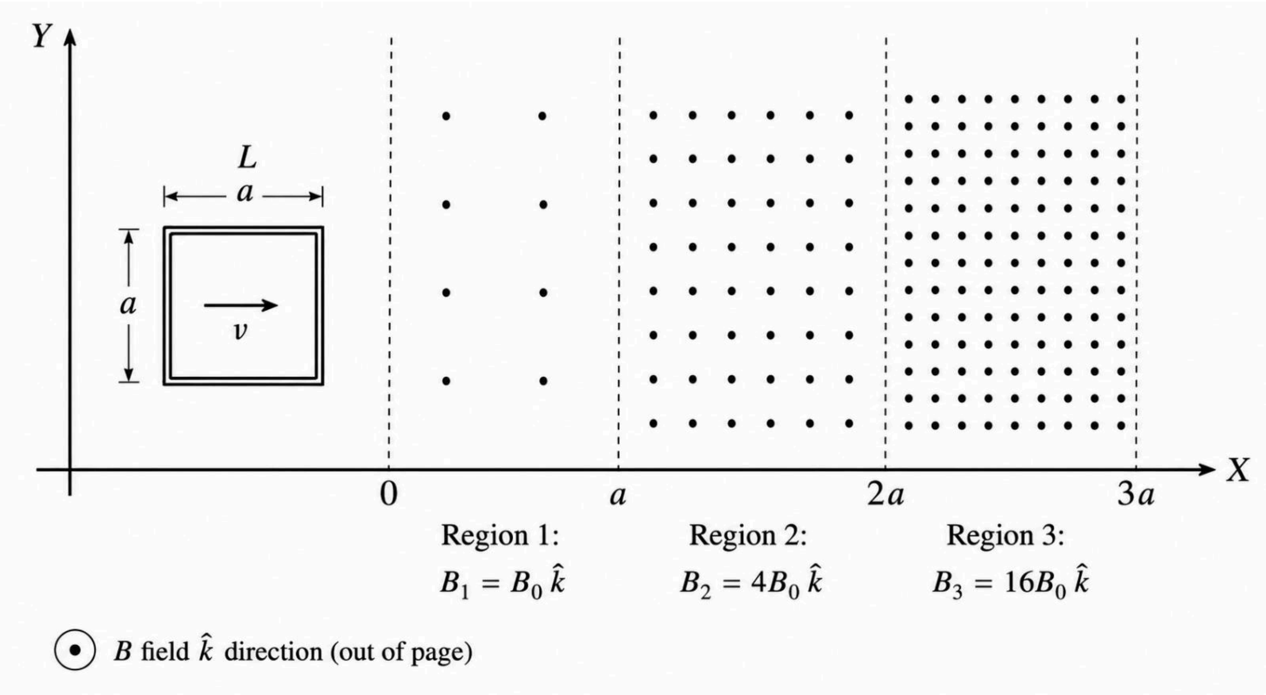
(A) The maximum number of different spectral lines observed in the emission spectrum is 1.	**(B)** The maximum number of different spectral lines observed in the emission spectrum is 6.
(C) The energy of the photon emitted in a direct transition to the ground state is 1.63×10^{-18} J.	**(D)** The wavelength of the highest-energy photon emitted is approximately 2.43×10^{-8} m.

Answer: B, D

Q.23 A rigid square loop of wire (labeled L) with side length a and resistance R is moving in the xy-plane with constant velocity $v = v\hat{i}$. It passes sequentially through three adjacent, non-overlapping regions (Region 1, Region 2, Region 3), each with a width of a (along the x-axis).

The magnetic fields in each region are constant and uniform, all directed perpendicularly out of the page ($+\hat{k}$ direction), with magnitudes as follows:
- Region 1 ($0 < x < a$): $B_1 = B_0\hat{k}$
- Region 2 ($a < x < 2a$): $B_2 = 4B_0\hat{k}$
- Region 3 ($2a < x < 3a$): $B_3 = 16B_0\hat{k}$

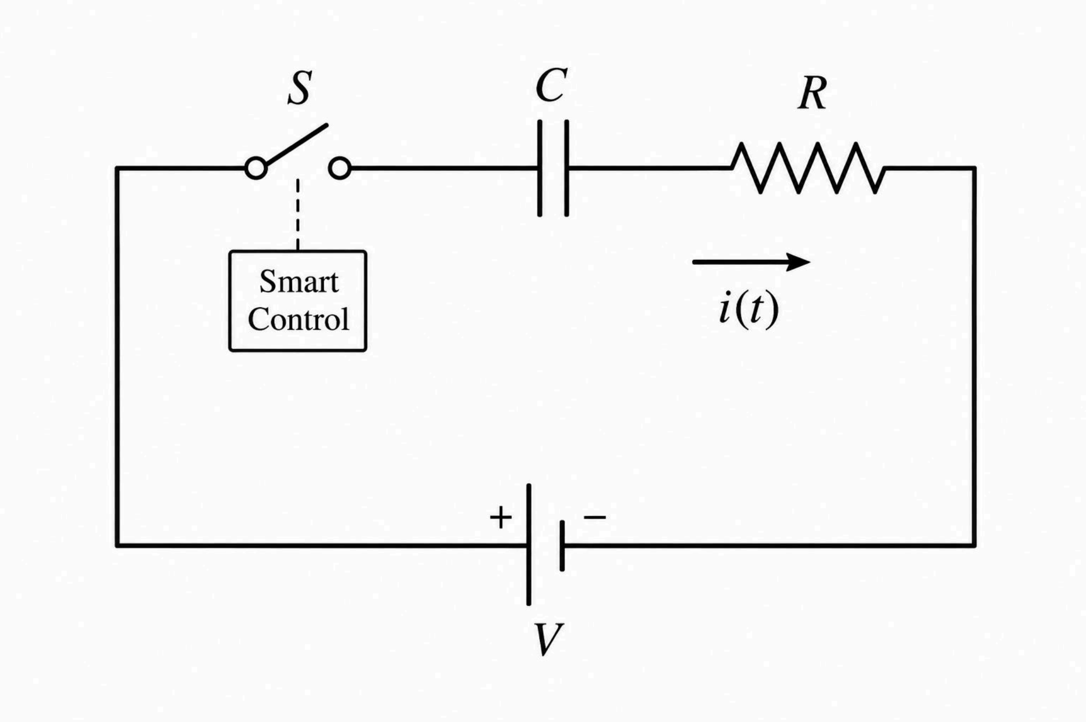
Outside these regions, the magnetic field is zero. Which of the following statements is/are correct regarding the loop's passage?



(A) When the loop is positioned such that its center is exactly at the boundary between Region 1 and Region 2 (i.e. $x = a$), the net magnetic force acting on the loop is $\frac{9a^2vB_0^2}{R}$ directed along the $+\hat{i}$ direction.	**(B)** The maximum mechanical power required from an external agent to maintain the loop's constant velocity occurs while it is completely exiting Region 3.
(C) The total mechanical work done by the external agent against magnetic forces during the loop's transition from Region 1 to Region 2 is equal to the total heat generated in the loop during the same transition interval.	**(D)** The ratio of total charge that flows through the loop while it passes from Region 2 to Region 3 to the total charge that flows while it enters Region 1 from the zero field region is $4:1$.

Answer: B, C

Q.24 An uncharged capacitor of capacitance C and a resistor of resistance R are connected in series to an ideal DC voltage source via a specialized sensor-switch S. At $t = 0$, the switch is closed, initiating an initial current I_0 in the circuit. The switch is programmed to automatically open (breaking the circuit) the exact moment the accumulated charge on the capacitor reaches a threshold value Q_{th}. Considering the system parameter $K = \frac{1}{RC}$, which of the following statements is/are correct?



(A) If $I_0 = 1.5KQ_{th}$, the charging current will naturally drop to zero before the switch opens.	**(B)** Once the capacitor charge reaches Q_{th}, the rate of change of charge in the circuit immediately becomes zero.
(C) If $I_0 = \frac{3}{4}KQ_{th}$, the current in the circuit becomes exactly zero at time $t = \left(\frac{1}{K}\right)\ln\left(\frac{4}{3}\right)$.	**(D)** If $I_0 = 2KQ_{th}$, the switch will automatically break the circuit at time $t = \left(\frac{1}{K}\right)\ln(2)$.

Answer: B, D

PHYSICS SECTION 3 (Maximum Marks : 24)

- This section contains **SIX (06)** questions.
- The answer to each question is a **NUMERICAL VALUE.**
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated **according to the following marking scheme:**

  Full Marks : +4 If **ONLY** the numerical answer is in acceptable range
  Zero Marks : 0 In all other cases

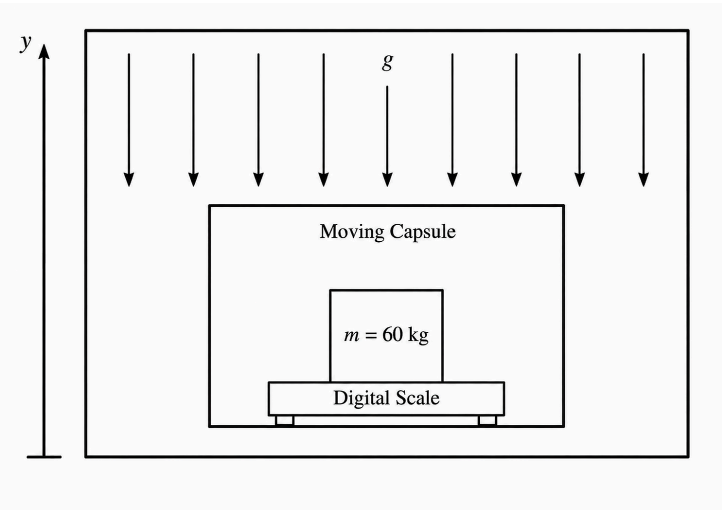
Q.25 A scientist is conducting a precision measurements test within a localized, uniform gravitational field generated by a specialized apparatus on a deep-space station. This apparatus creates a constant downward gravitational field g = 12 m/s² over the working area. A standard mass of 60 kg is placed on a high-precision digital scale resting on the floor of a testing capsule.

The capsule is programmed to move purely vertically within this field. Suppose the variation of the capsule's height, y (in cm), relative to the floor of the apparatus, varies with time t (in s) as:

$$y = 16\left[1 + \sin\left(\frac{\pi t}{3}\right)\right]$$

Taking $\pi^2 = 10$, what is the maximum difference between the object's recorded weight on the moving scale and its stationary true weight?

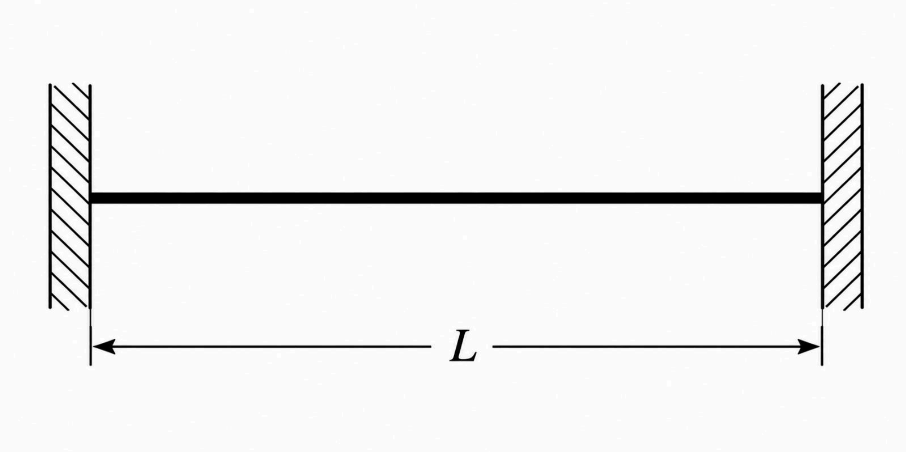
(Enter your answer as a numerical value rounded off to two decimal places.)



Answer: [10.6 to 10.7]

Q.26 A uniform cylindrical metallic wire of length $1.4 \mathrm{~m}$ and total mass $15.4 \times 10^{-2} \mathrm{~kg}$ is rigidly clamped between two immovable supports while at a uniform temperature of $55^{\circ} \mathrm{C}$. The wire is completely straight with zero initial tension. The wire is then uniformly cooled down to $20^{\circ} \mathrm{C}$. The radius of the cross-section of the wire is $1.0 \times 10^{-3} \mathrm{~m}$, its Young's modulus is $2.0 \times 10^{11} \mathrm{~N} / \mathrm{m}^{2}$, and its coefficient of linear thermal expansion is $1.8 \times 10^{-5} \mathrm{~K}^{-1}$.

If the wire is plucked in the middle, it vibrates in its fundamental mode. Taking $\pi=\frac{22}{7}$, the fundamental frequency of the resulting transverse standing waves in Hz (rounded off to two decimal places) is _____

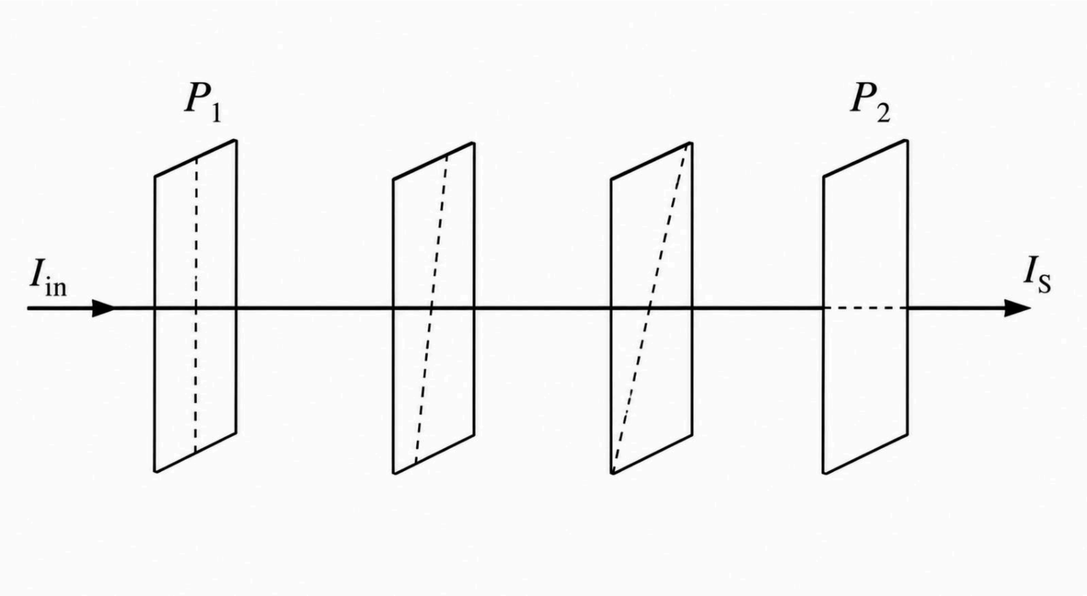


Answer: [21.4 to 21.5]

Q.27 Two ideal polarizing sheets, P_1 and P_2, are initially placed such that unpolarized light of intensity I_{in} incident on P_1 results in zero transmitted intensity through P_2. Let I_{max} be the maximum possible transmitted intensity through this initial two-sheet system (achieved if P_2 were to be rotated to be parallel to P_1).

Subsequently, two more identical ideal polarizing sheets are introduced between P_1 and P_2. The transmission axes of all four sheets are arranged such that the angle between the transmission axes of any two adjacent sheets is identical. If the final transmitted intensity in this new four-sheet setup is I_S, calculate the value of the ratio $\frac{I_{max}}{I_S}$.

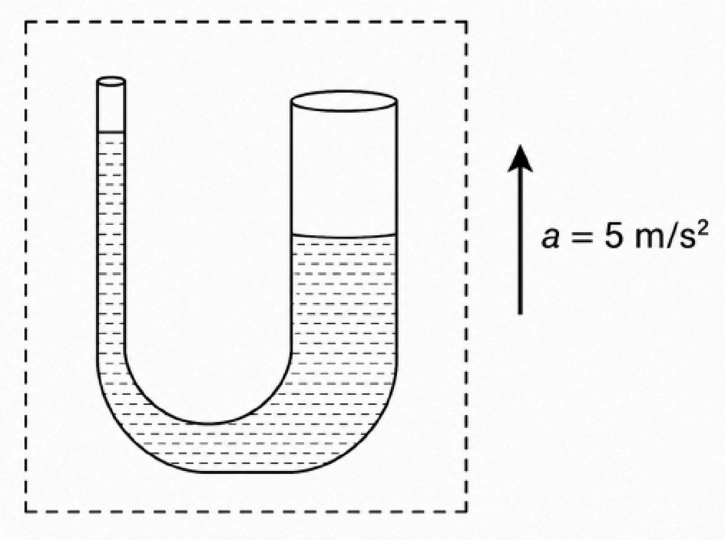
(Round off your answer to two decimal places)



Answer: [2.3 to 2.4]

Q.28 A U-tube with arm radii 0.2 mm and 0.4 mm contains water ($S = 0.07$ N/m, density $= 1000\ \mathrm{kg/m^3}$, contact angle 0°). Placed in an elevator accelerating upwards at $5\ \mathrm{m/s^2}$, find the steady difference in water levels between the arms in mm. (Round off your answer to two decimal places)

(Use $g = 10\ \mathrm{m/s^2}$)

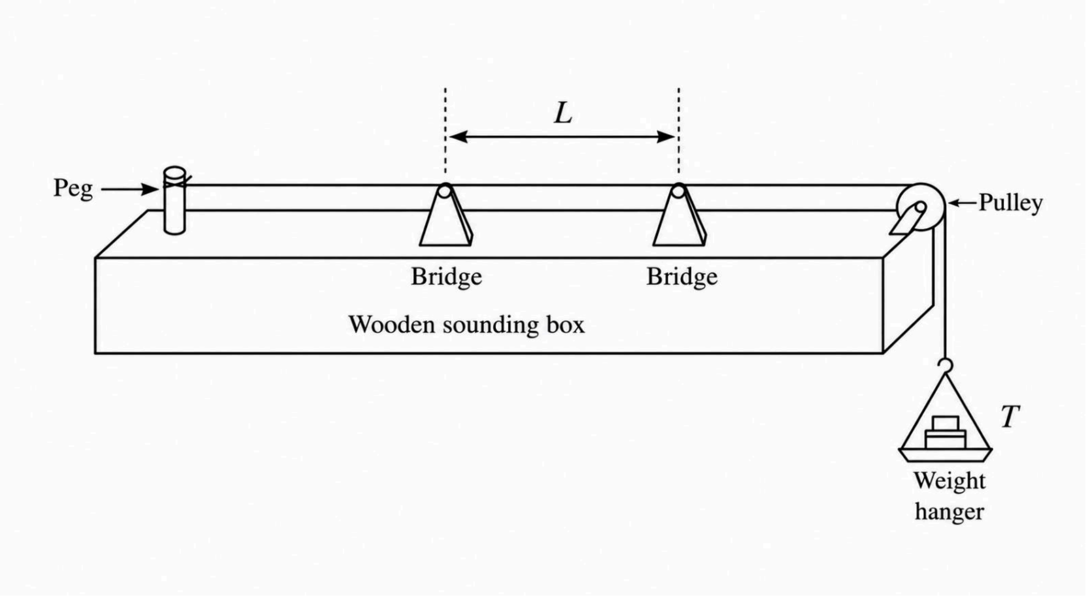


Answer: [23.3 to 23.4]

Q.29 A sonometer experiment is performed to determine the linear mass density of a wire using the equation $\mu = \dfrac{p^2 T}{4 f^2 L^2}$, where μ is the linear mass density, p is the number of antinodes formed, T is the tension applied to the wire, f is the frequency of vibration, and L is the resonant length of the vibrating segment. The tension T is applied by suspending weights (as shown in the figure) and is determined with a maximum absolute error of $500\ \mathrm{mN}$. The resonant length L is measured with a maximum absolute error of $1\ \mathrm{mm}$. The values of f and p are known precisely to be $50\ \mathrm{Hz}$ and 2, respectively.

Calculate the absolute error (in $\mathrm{g/m}$) in the value of μ estimated using the resonance state that occurs for $L = 60\ \mathrm{cm}$ and $T = 27\ \mathrm{N}$.

(Express your answer as a decimal rounded off to two decimal places).



Answer: [0.6 to 0.7]

Q.30 Consider n moles of an ideal monoatomic gas undergoing an adiabatic expansion where its temperature decreases by a positive amount ΔT. The work done by the gas during this process is found to be exactly equal to the energy stored in a massless spring of force constant k when it is compressed by a distance x.

A single photon has an energy exactly equal to the work done by this gas, and its wavelength λ is observed to be exactly equal to the spring's compression x.

If the drop in temperature of the gas is given by $\Delta T = \left(\dfrac{4h^2c^2k}{\alpha n^3 R^3}\right)^{1/3}$, (where h is Planck's constant, c is the speed of light in vacuum, and R is the universal gas constant) then the value of the dimensionless constant α is _____.

Answer: 27

PHYSICS SECTION 4 (Maximum Marks : 12)

- This section contains **FOUR (04)** Matching List Sets.
- Each set has **ONE** Multiple Choice Question.
- Each set has **TWO** lists: **List-I** and **List-II**.
- **List-I** has **Four** entries (P), (Q), (R) and (S) and **List-II** has **Five** entries (1), (2), (3), (4) and (5).
- **FOUR** options are given in each Multiple Choice Question based on **List-I** and **List-II** and **ONLY ONE** of these four options satisfies the condition asked in the Multiple Choice Question.
- Answer to each question will be evaluated **according to the following marking scheme:**

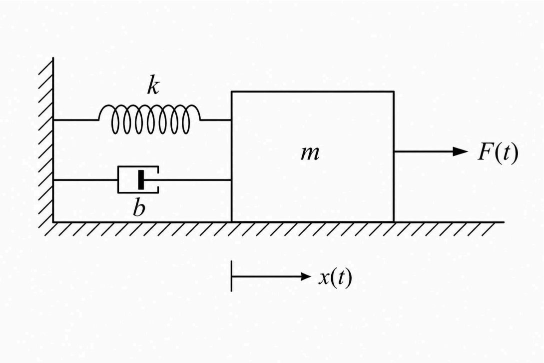
  Full Marks : +3 **ONLY** if the option corresponding to the correct combination is chosen;

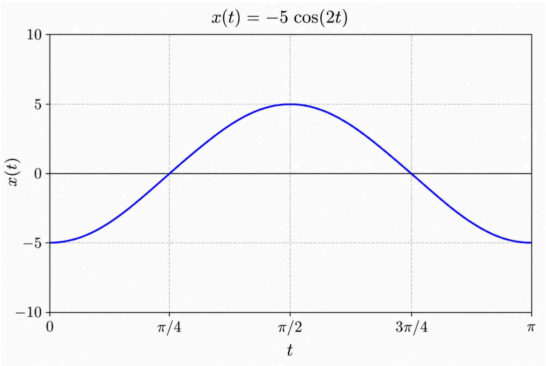
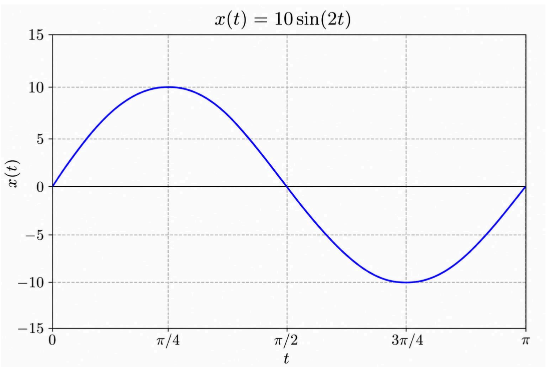
  Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);

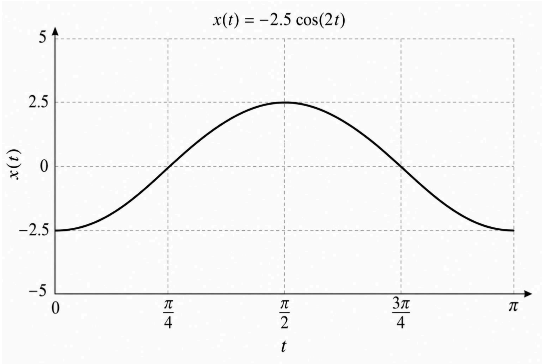
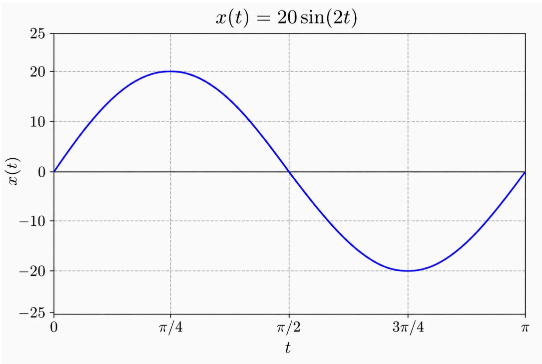
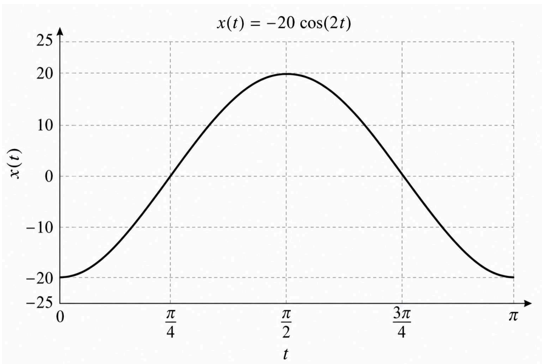
  Negative Marks : -1 In all other cases

Q.31 A mass-spring-damper system is subjected to an external driving force $F(t) = 20\sin(2t)$ N, where t is time in seconds. The steady-state displacement of the mass is given by $x(t) = A\sin(2t - \phi)$. List-I describes various system configurations with mass (m), spring constant (k), and damping coefficient (b). List-II shows possible displacement-time graphs $x(t)$.

Note: The amplitude is $A = \dfrac{F_0}{\sqrt{(k - m\omega^2)^2 + (b\omega)^2}}$ and phase lag is $\tan\phi = \dfrac{b\omega}{k - m\omega^2}$.



List-I (System Parameters)	List-II (Displacement Graphs)
(P) $m = 1\ \text{kg}, k = 5\ \text{N/m}, b = 0\ \text{kg/s}$	 (1)
(Q) $m = 1\ \text{kg}, k = 4\ \text{N/m}, b = 2\ \text{kg/s}$	 (2)

(R) $m = 1\ \mathrm{kg}, k = 4\ \mathrm{N/m}, b = 4\ \mathrm{kg/s}$	(3) 
(S) $m = 2\ \mathrm{kg}, k = 10\ \mathrm{N/m}, b = 0\ \mathrm{kg/s}$	(4) 
	(5) 

Choose the correct matching:

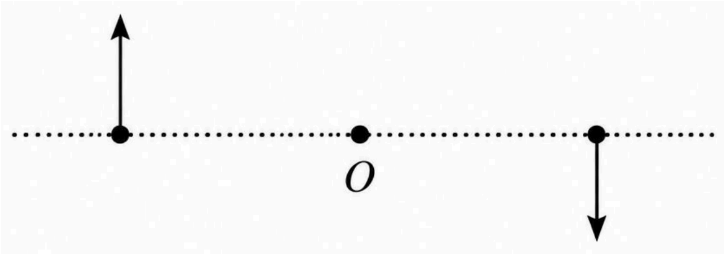
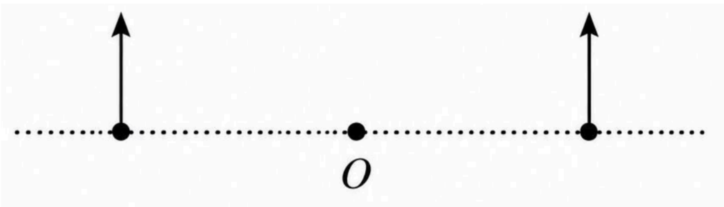
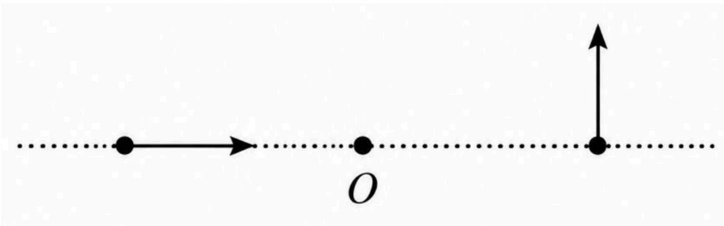
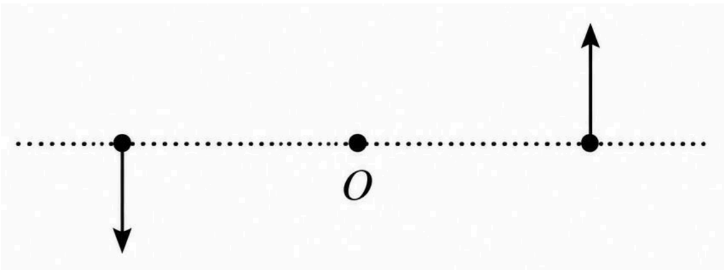
(A) $P \rightarrow 4, Q \rightarrow 5, R \rightarrow 1, S \rightarrow 5$	**(B)** $P \rightarrow 4, Q \rightarrow 1, R \rightarrow 3, S \rightarrow 2$
(C) $P \rightarrow 2, Q \rightarrow 5, R \rightarrow 4, S \rightarrow 3$	**(D)** $P \rightarrow 2, Q \rightarrow 3, R \rightarrow 4, S \rightarrow 1$

Answer: B

Q.32 List-I shows four configurations, each consisting of a rigid, massless rod of length $2L$ placed along the x-axis. The center of each rod is fixed at the origin $O(0,0)$. In each configuration, two forces, each of magnitude F, are applied to the ends of the rod at coordinates $x=-L$ and $x=+L$. The directions of these forces are indicated by the arrows in the figures. The possible resultant net torque τ about the origin O is given in List-II.

Choose the option that describes the correct match between the entries in **List-I** to those in **List-II**.

(Assume standard xy-coordinate plane where $\hat{\imath}$ is right, $\hat{\jmath}$ is up, and $\hat{k}$ is out of the page)

List-I (Force Configurations)	List-II (Resultant Net Torque τ)
(P) 	(1) $\tau=0$
(Q) 	(2) $\tau=FL\hat{k}$
(R) 	(3) $\tau=-2FL\hat{k}$
(S) 	(4) $\tau=2FL\hat{k}$
	(5) $\tau=-FL\hat{k}$

(A) $P\rightarrow 4, Q\rightarrow 2, R\rightarrow 5, S\rightarrow 3$	**(B)** $P\rightarrow 3, Q\rightarrow 1, R\rightarrow 2, S\rightarrow 4$
(C) $P\rightarrow 3, Q\rightarrow 2, R\rightarrow 1, S\rightarrow 4$	**(D)** $P\rightarrow 4, Q\rightarrow 1, R\rightarrow 2, S\rightarrow 3$

Answer: B

Q.33 List-I shows various functional dependencies of a physical parameter (y) on the wavelength (λ) of electromagnetic radiation in a vacuum. Parameters associated with certain physical phenomena or components are given in List-II.

Choose the option that describes the correct match between the entries in List-I to those in List-II.

List-I	List-II
(P) $y \propto \lambda^{-4}$	(1) magnitude of the momentum of a single photon of the wave
(Q) $y \propto \lambda^{-1}$	(2) maximum stopping potential in a photoelectric experiment using the wave
(R) $y \propto \lambda$	(3) intensity of the wave after undergoing Rayleigh scattering in the atmosphere
(S) y is practically independent of λ	(4) ratio of the electric field amplitude to the magnetic field amplitude in a vacuum
	(5) fringe width produced by the wave in a standard Young's double-slit apparatus

(A) $P \to 3, Q \to 2, R \to 1, S \to 5$	**(B)** $P \to 1, Q \to 3, R \to 2, S \to 4$
(C) $P \to 3, Q \to 1, R \to 5, S \to 4$	**(D)** $P \to 3, Q \to 2, R \to 4, S \to 5$

Answer: C

Q.34 An alternating voltage $V = V_0 \sin(\omega t)$ is applied to different electrical circuit configurations described in Column I. Match each circuit configuration with its corresponding steady-state characteristics given in Column II, and choose the correct option.

Column I	Column II
(P) A series L-R circuit having an inductive reactance equal to its resistance.	(1) The power factor of the circuit is $\frac{1}{\sqrt{2}}$.
(Q) A series C-R circuit having a capacitive reactance equal to $\sqrt{3}$ times its resistance.	(2) The power factor of the circuit is 0.5.
(R) A series L-C-R circuit having an inductive reactance twice its capacitive reactance, and resistance equal to its capacitive reactance.	(3) The main current lags the applied voltage by a phase angle of $\frac{\pi}{4}$.
(S) A parallel combination of an ideal inductor and an ideal capacitor having a capacitive reactance twice its inductive reactance.	(4) The main current leads the applied voltage by a phase angle of $\frac{\pi}{3}$.
	(5) The main current lags the applied voltage by a phase angle of $\frac{\pi}{2}$.

(A) $\mathrm{P} \rightarrow 1,3;\ \mathrm{Q} \rightarrow 2,4;\ \mathrm{R} \rightarrow 1,3;\ \mathrm{S} \rightarrow 5$	**(B)** $\mathrm{P} \rightarrow 1;\ \mathrm{Q} \rightarrow 2,3;\ \mathrm{R} \rightarrow 3,5;\ \mathrm{S} \rightarrow 2,5$
(C) $\mathrm{P} \rightarrow 1,3;\ \mathrm{Q} \rightarrow 2,4;\ \mathrm{R} \rightarrow 1,3;\ \mathrm{S} \rightarrow 2,5$	**(D)** $\mathrm{P} \rightarrow 3;\ \mathrm{Q} \rightarrow 2,3;\ \mathrm{R} \rightarrow 3,4;\ \mathrm{S} \rightarrow 2,5$

Answer: A

CHEMISTRY SECTION 1 (Maximum Marks : 12)

- This section contains **FOUR (04)** questions.
- Each question has **FOUR options** (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated **according to the following marking scheme:**

  Full Marks : +3 If **ONLY** the correct option is chosen
  Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
  Negative Marks : -1 In all other cases.

Q.35 Consider the structural characteristics of the disaccharides Sucrose and Lactose. Which of the following options correctly pairs both the constituent monosaccharides and the exact nature of their glycosidic linkage?

(A) **Sucrose:** α-D-Glucose and β-D-Fructose ($C_1 - C_4$ linkage) **Lactose:** β-D-Galactose and β-D-Glucose ($C_1 - C_4$ linkage)	**(B)** **Sucrose:** α-D-Glucose and β-D-Fructose ($C_1 - C_2$ linkage) **Lactose:** α-D-Galactose and β-D-Glucose ($C_1 - C_4$ linkage)
(C) **Sucrose:** α-D-Glucose and β-D-Fructose ($C_1 - C_2$ linkage) **Lactose:** β-D-Galactose and β-D-Glucose ($C_1 - C_4$ linkage)	**(D)** **Sucrose:** β-D-Glucose and α-D-Fructose ($C_1 - C_2$ linkage) **Lactose:** β-D-Galactose and α-D-Glucose ($C_1 - C_2$ linkage)

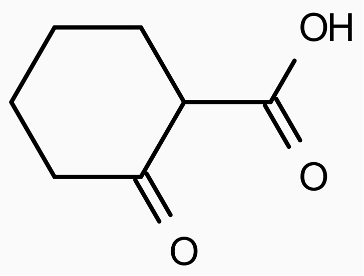
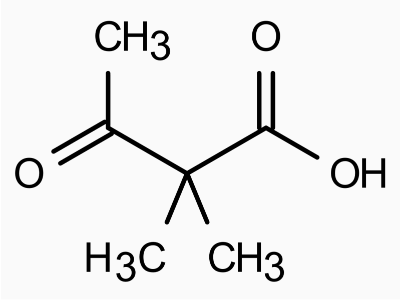
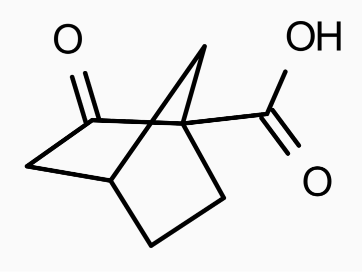
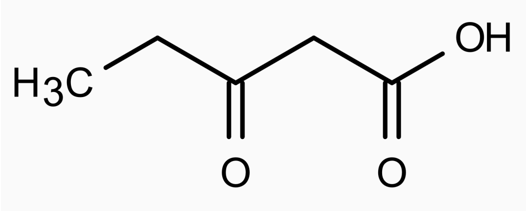
Answer: C

Q.36 A closed container holds a mixture of an ideal monatomic gas A and an ideal rigid diatomic gas B at a constant equilibrium temperature. The ratio of the total internal energy of gas A to that of gas B in the mixture is $1:2$. The mixture is allowed to effuse through a tiny pinhole into a vacuum. If the initial mass rate of effusion of gas A is exactly equal to the initial mass rate of effusion of gas B, what is the ratio of their molar masses $(M_A : M_B)$?

(A) $25:36$	**(B)** $36:25$
(C) $1:4$	**(D)** $4:1$

Answer: B

Q.37 Four different β-keto carboxylic acids are heated to the same temperature ($150^\circ\mathrm{C}$) to undergo thermal decarboxylation. Which of the following compounds will undergo decarboxylation at the **slowest rate** (or completely fail to react)?

(A) 	**(B)** 
(C) 	**(D)** 

Answer: C

Q.38 The saturated macromolecule formed by the exhaustive catalytic hydrogenation of polystyrene is structurally identical to the homopolymer of:

(A) vinylcyclohexane	**(B)** allylcyclohexane
(C) 1-cyclohexylpropene	**(D)** cyclohexene

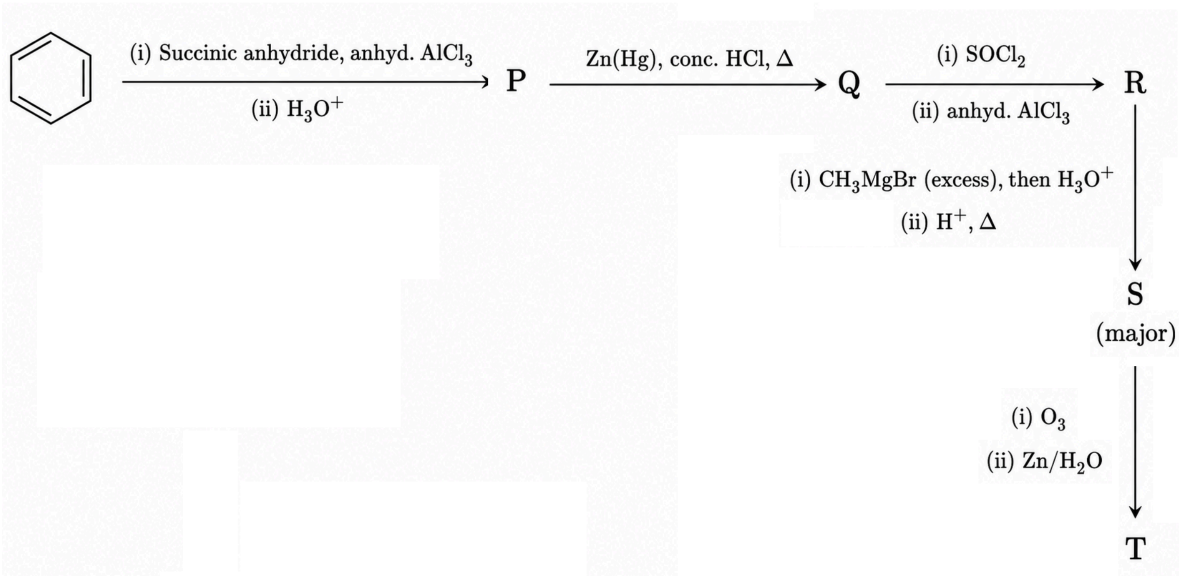
Answer: A

CHEMISTRY SECTION 2 (Maximum Marks : 12)

- This section contains **THREE (03)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated **according to the following marking scheme:**

  Full Marks : +4 **ONLY** if (all) the correct option(s) is(are) chosen;
  Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen;
  Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct;
  Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option;
  Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
  Negative Marks : -2 In all other cases.

Q.39 Consider the following sequence of reactions:



Which of the following statement(s) is/are **correct** regarding the given reaction sequence and its products?

(A) Compound T gives both a silver mirror with Tollens' reagent and a yellow precipitate with $\mathrm{I_2/NaOH}$.	**(B)** Upon treatment with dilute NaOH followed by heating, T predominantly undergoes intramolecular aldol condensation to form a bicyclic compound containing a newly formed five-membered ring.
(C) Compound R is an aromatic ketone that fails to undergo the haloform reaction, whereas compound P is a keto-acid.	**(D)** Ozonolysis of compound S followed by oxidative workup $(\mathrm{H_2O_2})$ yields a dicarboxylic acid as the major organic product.

Answer: A, B, C

Q.40 Consider the principles of Molecular Orbital Theory (MOT) applied to diatomic molecules and their constituent atoms. Which of the following statements is/are correct?

(A) The transformation of O_2 into the O_2^+ ion increases the bond order and decreases the bond length, while the species remains paramagnetic.	**(B)** According to MOT, the ground state C_2 molecule is diamagnetic and possesses a double bond consisting of one σ-bond and one π-bond.
(C) The first ionization enthalpy of the N_2 molecule is higher than that of atomic nitrogen, whereas the first ionization enthalpy of the O_2 molecule is lower than that of atomic oxygen.	**(D)** The isoelectronic species NO^+ and CN^- are both diamagnetic, and their highest occupied molecular orbital (HOMO) possesses π symmetry.

Answer: A, C

Q.41 An inorganic salt M dissolves in water to form a solution. When aqueous $NaOH$ is added to this solution, a green precipitate W is formed. The precipitate W dissolves upon the addition of excess aqueous $NaOH$ and H_2O_2, yielding a yellow solution X. Separately, when the solid salt M is heated with solid $K_2Cr_2O_7$ and concentrated H_2SO_4, a reddish-brown gas Y is evolved. Passing gas Y into an aqueous solution of $NaOH$ produces a yellow solution Z.

Which of the following statements is/are CORRECT?

(A) The yellow solution X and the yellow solution Z contain the same coloured anionic species.	**(B)** The reddish-brown gas Y is a mixed anhydride of two inorganic acids and is paramagnetic in nature.
(C) Acidification of the yellow solution Z with dilute H_2SO_4 followed by the addition of H_2O_2 in the presence of amyl alcohol produces a blue coloured organic layer.	**(D)** The salt M reacts with concentrated H_2SO_4 on mild heating to produce a pungent-smelling colourless gas that gives dense white fumes with a glass rod dipped in aqueous ammonia.

Answer: A, C, D

CHEMISTRY SECTION 3 (Maximum Marks : 24)

- This section contains **SIX (06)** questions.
- The answer to each question is a **NUMERICAL VALUE.**
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated **according to the following marking scheme:**

  Full Marks : +4 If **ONLY** the numerical answer is in acceptable range
  Zero Marks : 0 In all other cases

Q.42 A hypothetical super-dense alloy crystallizes in a face-centered cubic (FCC) lattice. The edge length of its unit cell is determined via X-ray diffraction to be $0.2\ \mathrm{nm}$. If the molar mass of this alloy is $120.44\ \mathrm{g\ mol^{-1}}$, the density of the crystal (in $\mathrm{g\ cm^{-3}}$) is _____.

Use: $N_A = 6.022 \times 10^{23}\ \mathrm{mol^{-1}}$

Answer: 100

Q.43 At $300\ \mathrm{K}$, the equilibrium vapor pressure of a pure liquid 'A' is $0.1\ \mathrm{bar}$. The standard enthalpy of vaporization ($\Delta H^\circ_{\mathrm{vap}}$) of the liquid at this temperature is $35.50\ \mathrm{kJ\ mol^{-1}}$. Calculate the standard entropy of vaporization ($\Delta S^\circ_{\mathrm{vap}}$) of liquid A at $300\ \mathrm{K}$ in $\mathrm{J\ K^{-1}\ mol^{-1}}$.

(Report your answer rounded off to two decimal places)

Use: $R = 8.314\ \mathrm{J\ K^{-1}\ mol^{-1}}$, $\ln(10) = 2.303$

Answer: [99 to 100]

Q.44 A univalent anionic surfactant ($\mathrm{Na^+A^-}$) in an aqueous solution has a Critical Micelle Concentration (CMC) of 4×10^{-3} M and an aggregation number of 60. At 300 K, an aqueous solution is prepared with a total initial surfactant concentration of 53×10^{-3} M. Assuming that all $\mathrm{A^-}$ ions above the CMC participate in micelle formation, all $\mathrm{Na^+}$ ions remain completely free in the solution (zero counter-ion binding), and the solution behaves ideally, calculate the osmotic pressure of this solution in atm.

(Use $R = 1/12$ L atm $\mathrm{K^{-1}\ mol^{-1}}$ and round off the final answer to two decimal places).

Answer: [1.4 to 1.5]

Q.45 A constant current of $7.0 \mathrm{~A}$ is passed through an electrolytic cell containing molten Al_2O_3. The electrolysis process produces $6.225 \mathrm{~L}$ of oxygen gas at the anode, measured at a pressure of $1 \mathrm{~bar}$ and a temperature of $300 \mathrm{~K}$. Assuming ideal gas behavior, calculate the time required for this process in **hours**.

(Round off the final answer to two decimal places)

Use: Faraday's constant (F) = $96500 \mathrm{~C~mol}^{-1}$
Universal gas constant (R) = $0.083 \mathrm{~L~bar~K}^{-1} \mathrm{~mol}^{-1}$

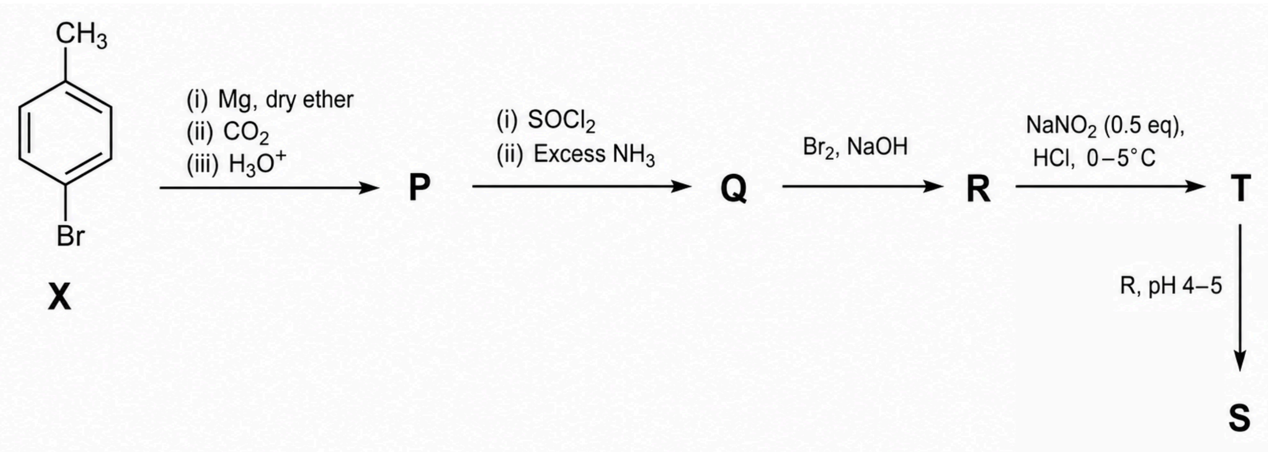
Answer: [3.75 to 3.95]

Q.46 An alcohol (X) with the molecular formula $C_4H_{10}O$ gives a blue color in the Victor Meyer test. When a certain mass, w grams, of (X) is treated with excess sodium metal, $448 \mathrm{~mL}$ of H_2 gas is evolved at STP ($1 \mathrm{~atm}$, $273 \mathrm{~K}$). If the exact same mass w of (X) is completely oxidized by an acidified $0.05 \mathrm{~M}$ $K_2Cr_2O_7$ solution, the volume (in mL) of the dichromate solution required is _____.

(Assume oxidation stops at the ketone stage. Use molar volume of an ideal gas at STP = $22.4 \mathrm{~L/mol}$*)*

Answer: [266.6 to 266.7]

Q.47 The reaction sequence given below is carried out with $500 \mathrm{~g}$ of **X**. Assume 100% conversion for the given stoichiometric equivalents. The amount (in grams) of **S** produced is _____. (Round off to 2 decimal places).



Use: Atomic mass (in amu): H = 1, C = 12, N = 14, O = 16, Br = 80

Answer: [328 to 330]

CHEMISTRY SECTION 4 (Maximum Marks : 12)

- This section contains **FOUR (04)** Matching List Sets.
- Each set has **ONE** Multiple Choice Question.
- Each set has **TWO** lists: **List-I** and **List-II**.
- **List-I** has **Four** entries (P), (Q), (R) and (S) and **List-II** has **Five** entries (1), (2), (3), (4) and (5).
- **FOUR** options are given in each Multiple Choice Question based on **List-I** and **List-II** and **ONLY ONE** of these four options satisfies the condition asked in the Multiple Choice Question.
- Answer to each question will be evaluated **according to the following marking scheme:**

  Full Marks : +3 **ONLY** if the option corresponding to the correct combination is chosen;
  Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
  Negative Marks : -1 In all other cases

Q.48 Match each coordination complex in Column I with its corresponding properties in Column II, and choose the correct option.

[Given: Atomic numbers of $\mathrm{Cr}=24, \mathrm{Co}=27, \mathrm{Ni}=28$]

Column I	Column II
(P) $[\mathrm{Co(en)_2Cl_2}]^+$	(1) Exhibits geometrical isomerism
(Q) $[\mathrm{Cr(NH_3)_5(NO_2)}]^{2+}$	(2) Central metal is d^2sp^3 hybridized
(R) $[\mathrm{Ni(en)_3}]^{2+}$	(3) Optically active stereoisomer exists
(S) $[\mathrm{Co(NH_3)_3(NO_2)_3}]$	(4) Paramagnetic with $\mu_{\text{spin-only}} > 2.0$ BM
	(5) Exhibits linkage isomerism

(A) $\mathrm{P} \rightarrow 1,2,3;\ \mathrm{Q} \rightarrow 2,4,5;\ \mathrm{R} \rightarrow 3,4;\ \mathrm{S} \rightarrow 1,2,5$	**(B)** $\mathrm{P} \rightarrow 1,2;\ \mathrm{Q} \rightarrow 2,4;\ \mathrm{R} \rightarrow 2,3,4;\ \mathrm{S} \rightarrow 2,4,5$
(C) $\mathrm{P} \rightarrow 1,2;\ \mathrm{Q} \rightarrow 2,4,5;\ \mathrm{R} \rightarrow 2,3;\ \mathrm{S} \rightarrow 3,5$	**(D)** $\mathrm{P} \rightarrow 1,2,3;\ \mathrm{Q} \rightarrow 2,4;\ \mathrm{R} \rightarrow 2,3,4;\ \mathrm{S} \rightarrow 2,5$

Answer: A

Q.49 Match the nitrogen-containing compounds in **List-I** with the most accurate description of their basicity or lone-pair electron behavior given in **List-II**.

List-I (Compound)	List-II (Characteristic / Behavior)
(P) Pyridine	(1) The lone pair is part of the aromatic sextet, making it a very weak base.
(Q) Pyrrole	(2) The conjugate acid is highly stabilized by equivalent resonance structures.
(R) Guanidine	(3) The lone pair resides in an sp^2 hybridized orbital, perpendicular to the π-system.
(S) Aniline	(4) The lone pair is delocalized over the benzene ring via the +R (mesomeric) effect.
	(5) Protonation of this species yields an anti-aromatic conjugate acid.

(A) $P \to 3; Q \to 5; R \to 1; S \to 4$	**(B)** $P \to 4; Q \to 5; R \to 2; S \to 3$
(C) $P \to 4; Q \to 1; R \to 3; S \to 5$	**(D)** $P \to 3; Q \to 1; R \to 2; S \to 4$

Answer: D

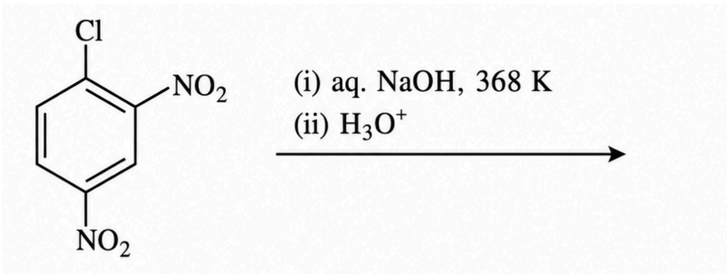
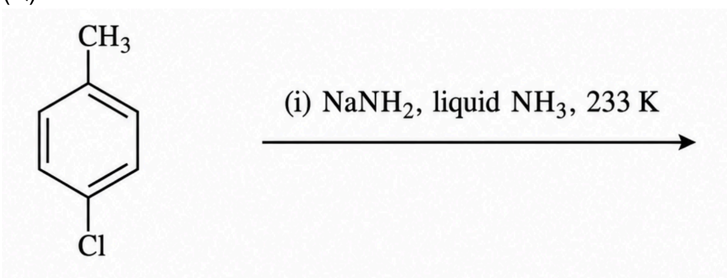
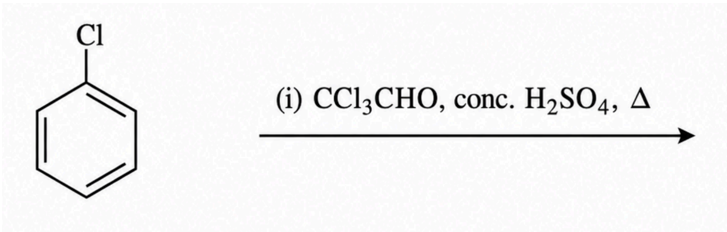
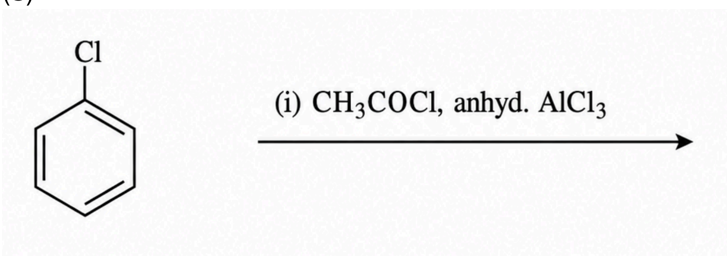
Q.50 Match the chemical transformations and properties in **Column I** with the appropriate observation or product in **Column II** and choose the correct option.

Column I	Column II
(P) Reaction of PCl_3 with C_2H_5OH (excess) $\rightarrow$	(1) Paramagnetic gas is evolved
(Q) Hydrolysis of $NCl_3 \rightarrow$	(2) Formation of a product containing $P-H$ bond
(R) Reaction of Cu with cold, dilute $HNO_3 \rightarrow$	(3) Formation of $HOCl$
(S) Solid PCl_5 reacts with water (limited) $\rightarrow$	(4) Product with tetrahedral geometry around central atom
	(5) Formation of N_2O gas

(A) $(P) \to 2,4;\ (Q) \to 3;\ (R) \to 1;\ (S) \to 4$	**(B)** $(P) \to 2;\ (Q) \to 3;\ (R) \to 5;\ (S) \to 4$
(C) $(P) \to 4;\ (Q) \to 1,3;\ (R) \to 5;\ (S) \to 2$	**(D)** $(P) \to 2,4;\ (Q) \to 1;\ (R) \to 3;\ (S) \to 5$

Answer: A

Q.51 Match the reactions/compounds in **List-I** with the appropriate mechanisms or observations in **List-II** and choose the correct option.

List-I	List-II
(P) 	(1) Proceeds via a highly reactive neutral intermediate; yields a mixture of structurally isomeric amine products.
(Q) 	(2) Reaction is an electrophilic aromatic substitution; the major product formed will give a positive yellow precipitate with alkaline I_2 (Iodoform test).
(R) 	(3) Reaction yields a highly substituted diphenyl trichloroethane derivative, known as a non-biodegradable insecticide.
(S) 	(4) Nucleophilic substitution occurs via a resonance-stabilized carbanion (Meisenheimer complex) due to the presence of strongly electron-withdrawing groups.
	(5) Proceeds via a stable aryl cation intermediate followed by rapid nucleophilic attack, yielding a single, pure substitution product.

(A) $P \to 4; Q \to 3; R \to 5; S \to 1$	**(B)** $P \to 4; Q \to 1; R \to 3; S \to 2$
(C) $P \to 5; Q \to 4; R \to 3; S \to 1$	**(D)** $P \to 4; Q \to 5; R \to 1; S \to 2$

Answer: B